

BALL AND PLAYERS TRACKING IN PADEL MATCHES VIDEOS

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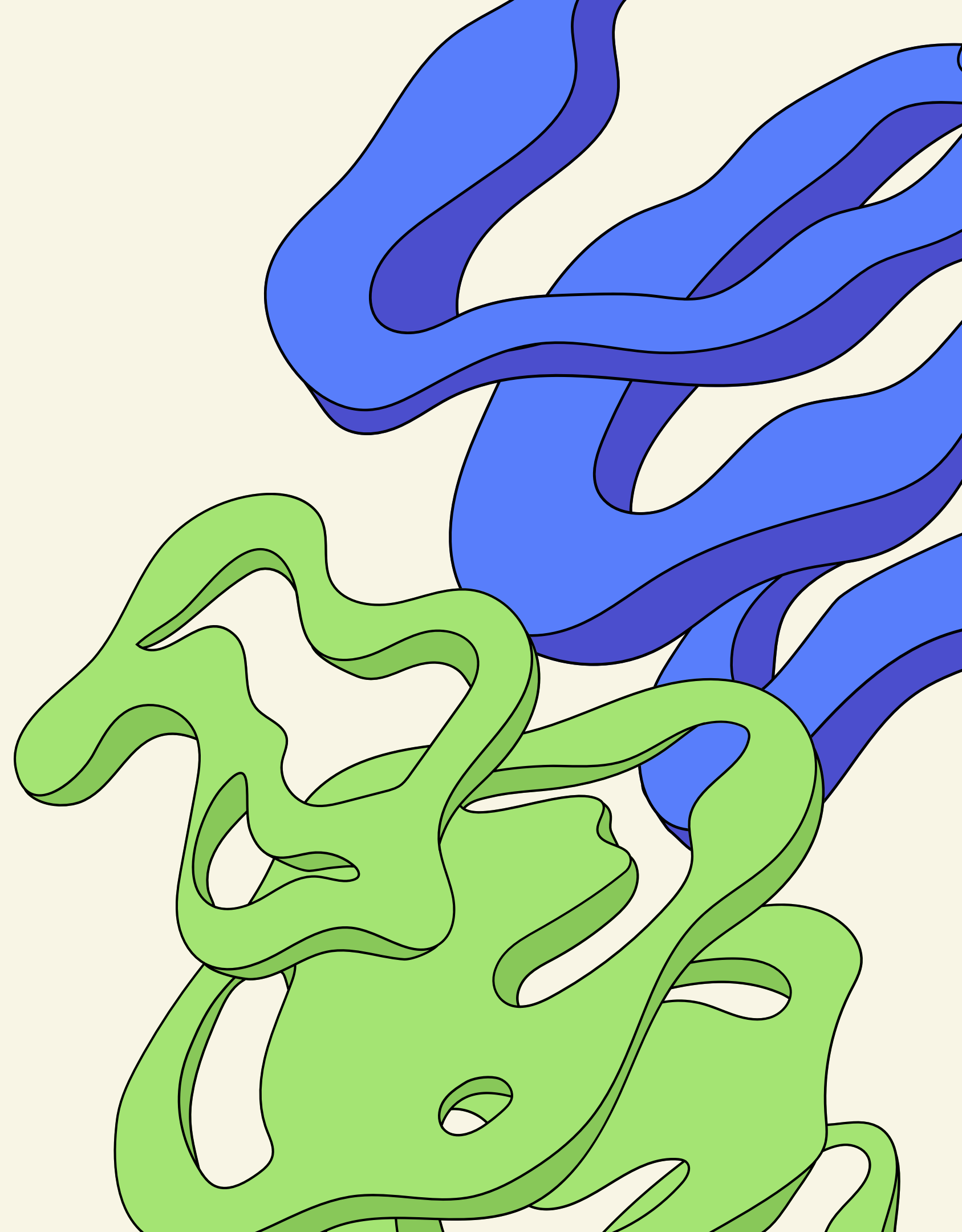
Davide Tonetti - s334297

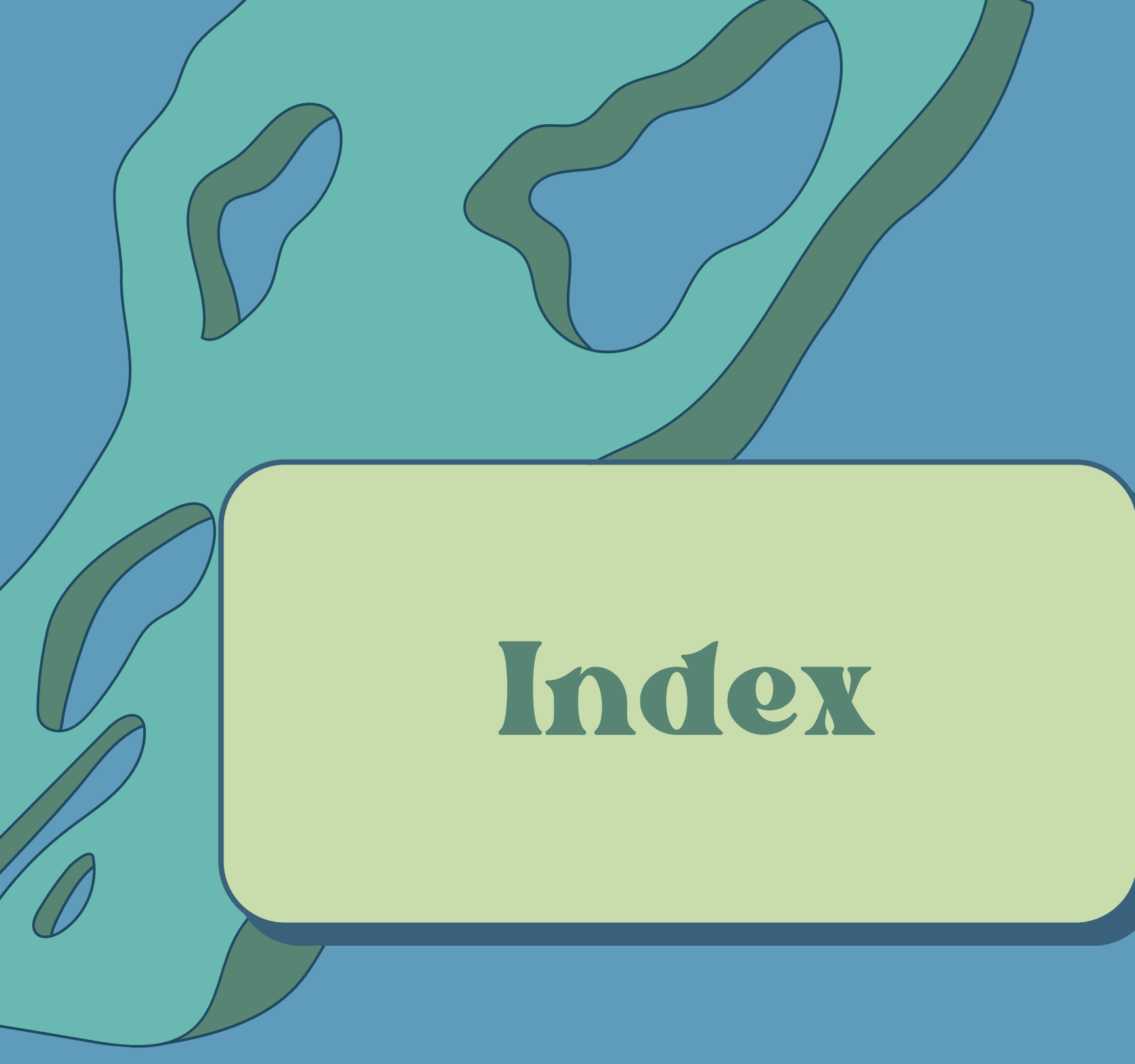
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links
PASSION FOR INNOVATION



A stylized map of the African continent is located in the top-left corner of the slide. It features a light teal background with darker teal and blue outlines representing the landmasses and surrounding water bodies.

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- **Pipeline**
- **Pre-trained models**
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- **Output statistics**
- **Conclusion**
- **Extras**

Objectives & Stakeholders



Project Objectives

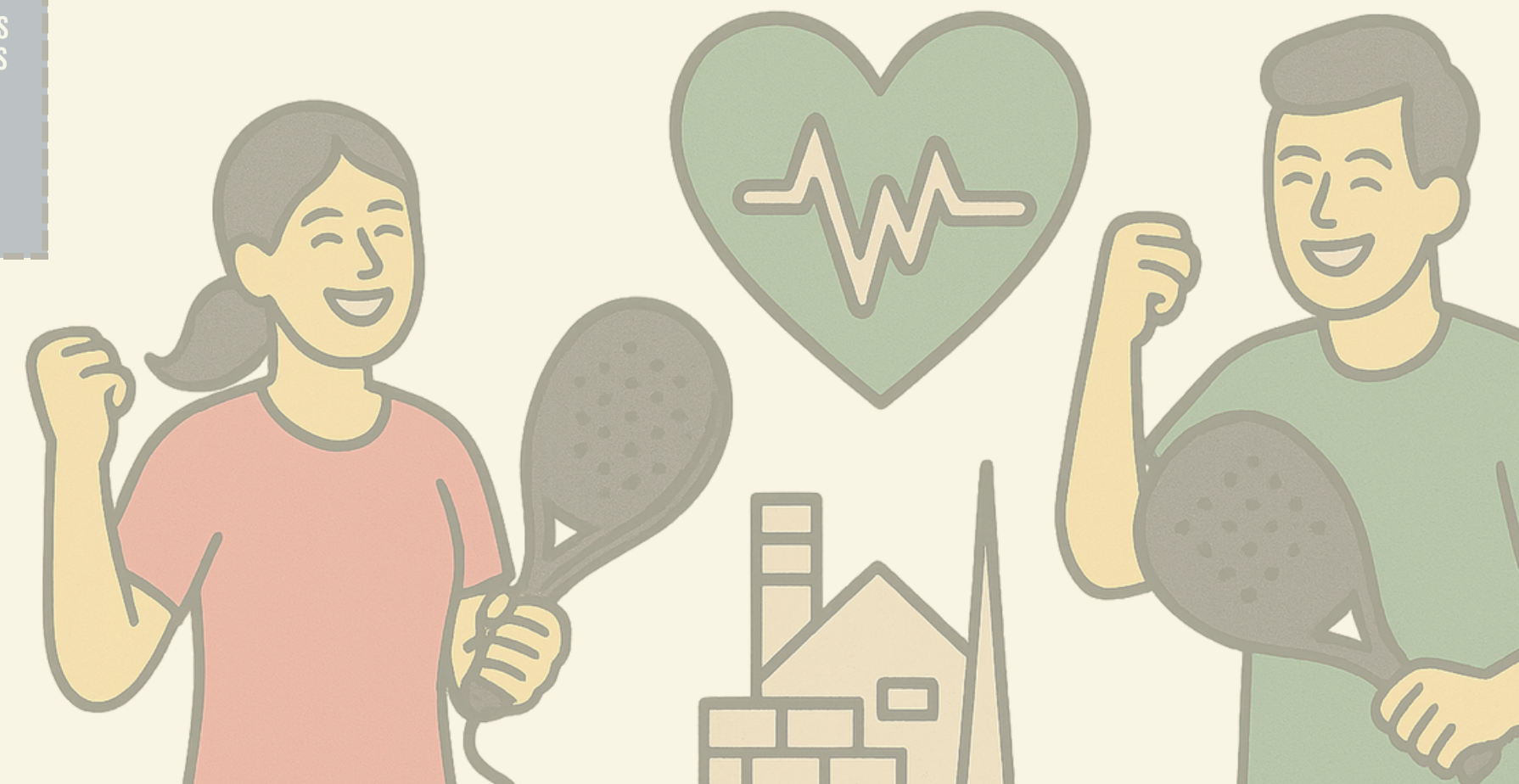
- **Objective and interpretable performance metrics**
- **Tactical analysis** (heatmaps, distances)
- Design a **scalable, data-driven pipeline** for single-camera setups



Target Stakeholders

- **Players:** improve awareness of positioning and movement patterns
- **Coaches:** support training decisions with quantitative insights
- **Performance Analysts:** automate video analysis and reduce manual effort

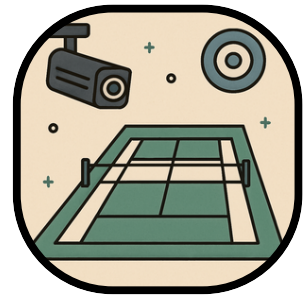
SDG Goals



Dataset



100,000 frames



**Single, fixed
camera angle**



**Frame-level
annotations**

zenodo

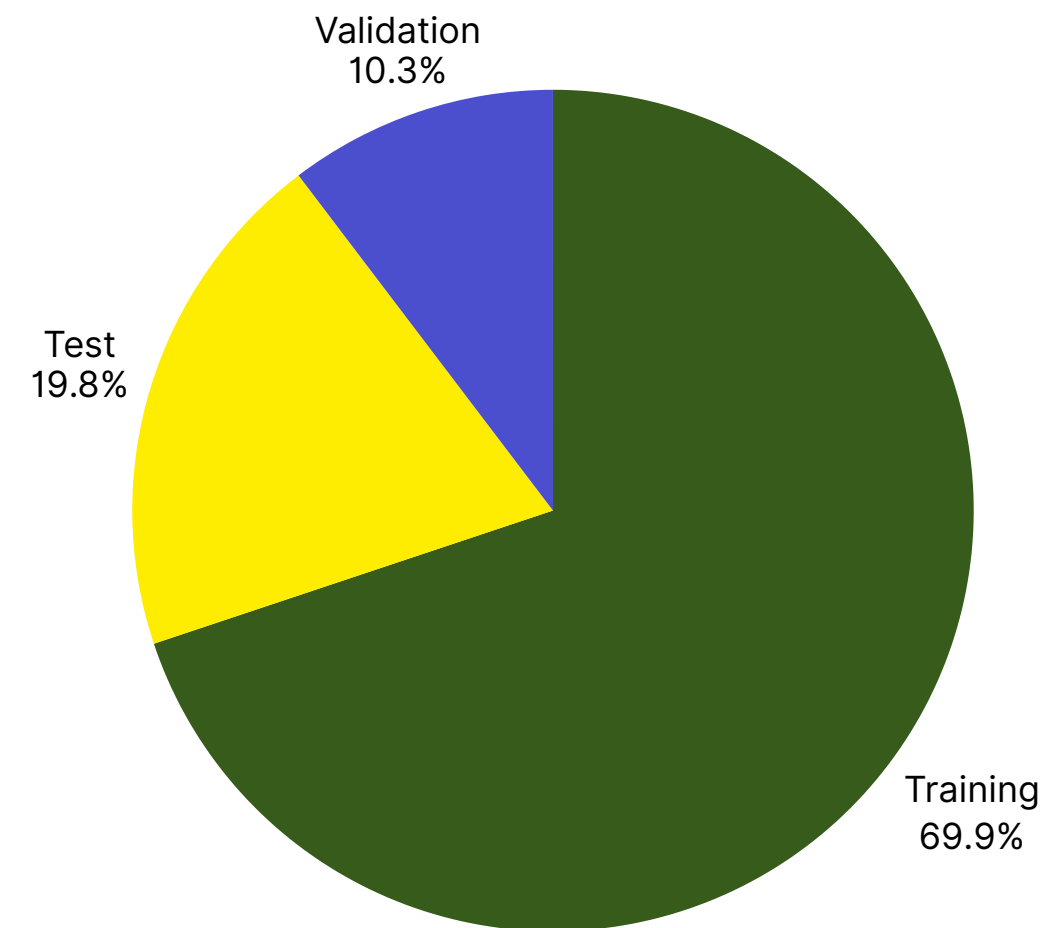
PadelTracker100: A Dataset for Intelligent Player and Ball Tracking in Padel Sports

Author: [Bada-Nerín, Roberto](#); [Paula Rodríguez](#); [Kreibel, Denis](#); [Teodoro Rodríguez, Joao](#); [Víctor Pedrayes, Oscar D.](#); [Usamentiaga, Ruben](#); [Fontenla-Seco, Yago](#); [Ben Lestón, Pablo](#); [Rodríguez Trillo, Sebastián](#); [Lozano García, Nicolás](#); [Mosquera Bonasorte, Franco](#)

Published: 2025

Source: Zenodo

Split Strategy



Hardware

Q: Where do we store and process this large amount of data?

Google Colab



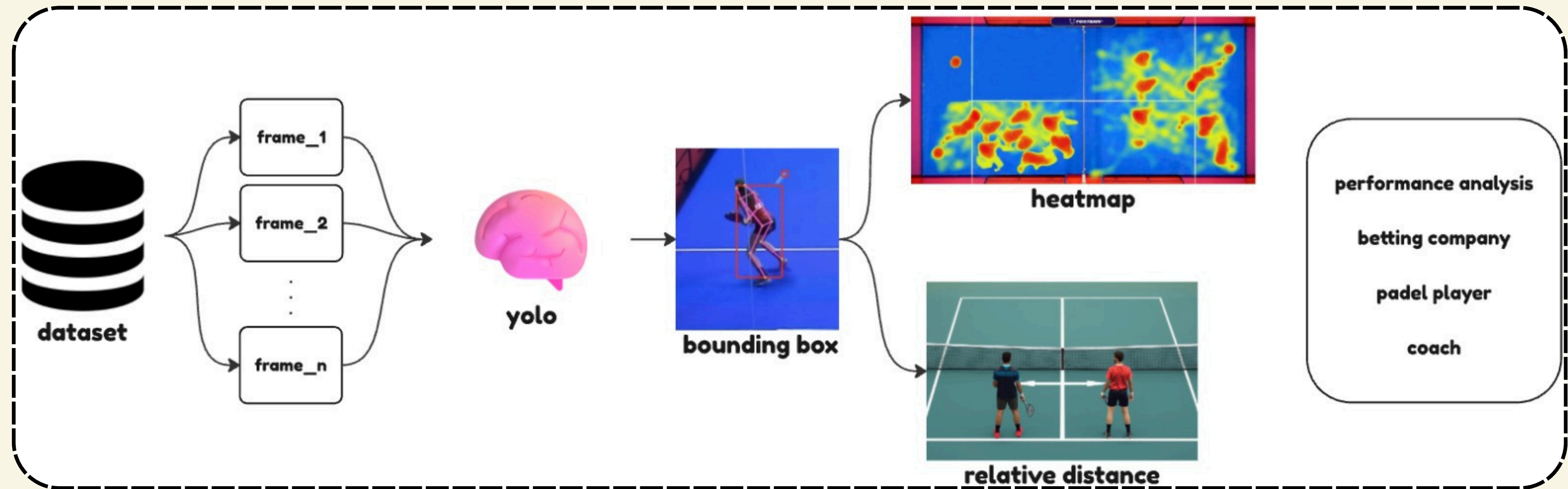
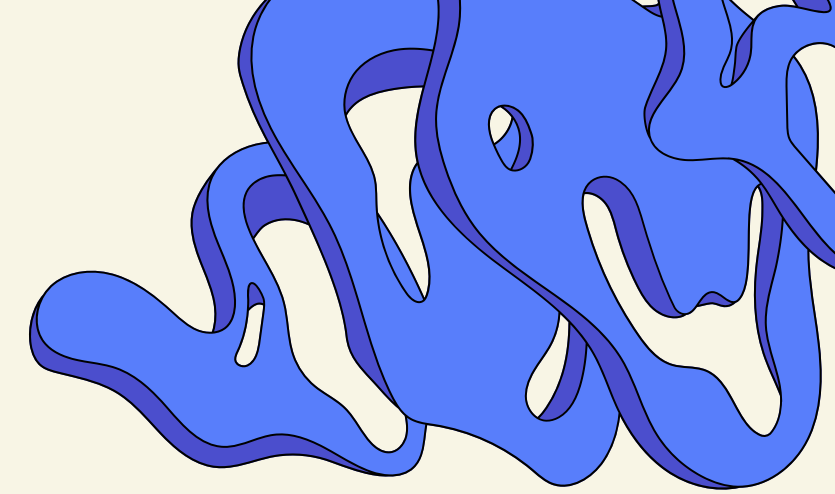
Limited disk storage capacity, GPU memory and runtime constraints

LINKS Asgard



Large scale storage, dedicated high-memory GPUs, stable environment

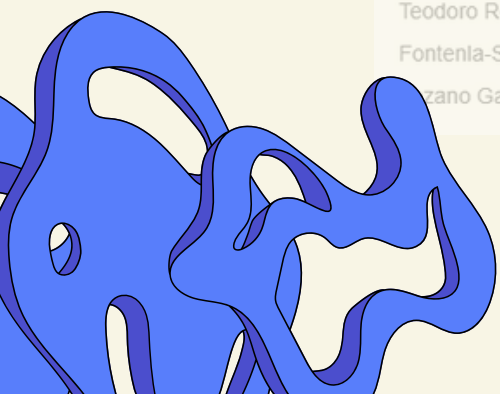
Pipeline



PadelTracker100: A Dataset for Intelligent Player and Ball Tracking in Padel Sports

Bada-Nerín, Roberto (Researcher)¹ ; Paula Rodríguez (Researcher)¹ ; Kreibel, Denis (Supervisor)¹ ;
Teodoro Rodríguez, Joao Víctor (Researcher)¹ ; Pedrayes, Oscar D. (Supervisor)¹ ; Usamentiaga, Ruben (Supervisor)² ;
Fontenla-Seco, Yago (Supervisor)¹ ; Ben Lestón, Pablo (Annotator)¹ ; Rodríguez Trillo, Sebastián (Annotator)¹ ;
Zano García, Nicolás (Annotator)¹ ; Mosquera Bonasorte, Franco (Researcher)

Show affiliations

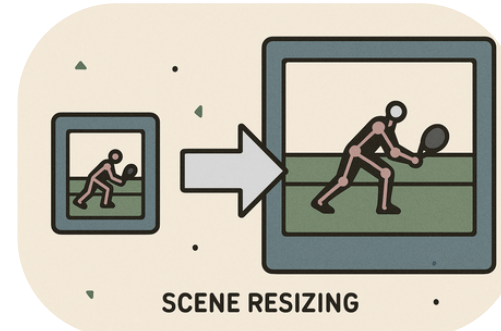


Models



yolo 11n
yolo 10n
yolo 9t

model comparison



640
960
1280

imgsz hyperparameter

Recall: how many true positives are detected among all actual positives

Precision: how many of the model's positive predictions are correct

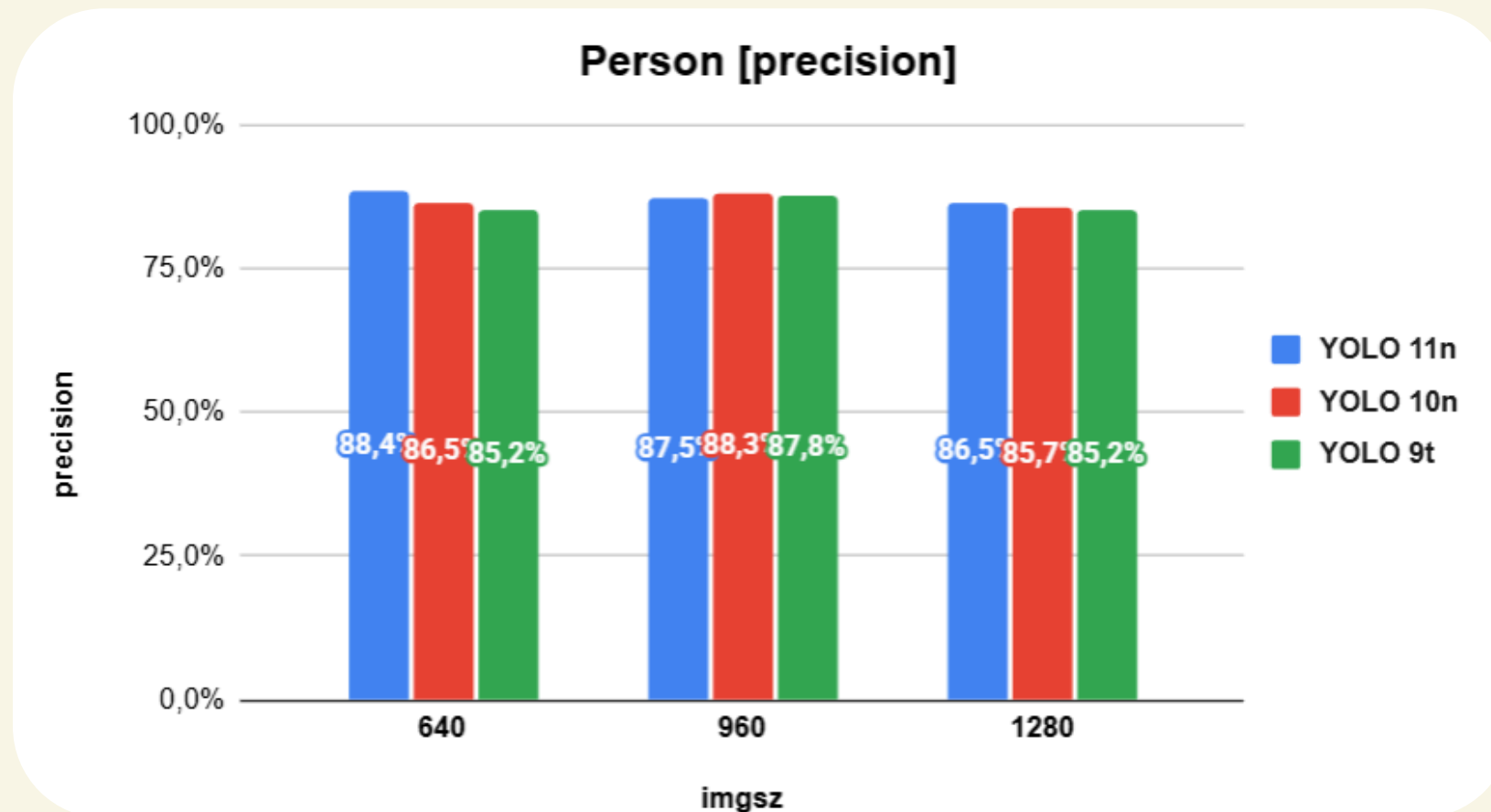
Metrics

1. Evaluate the pre-trained models

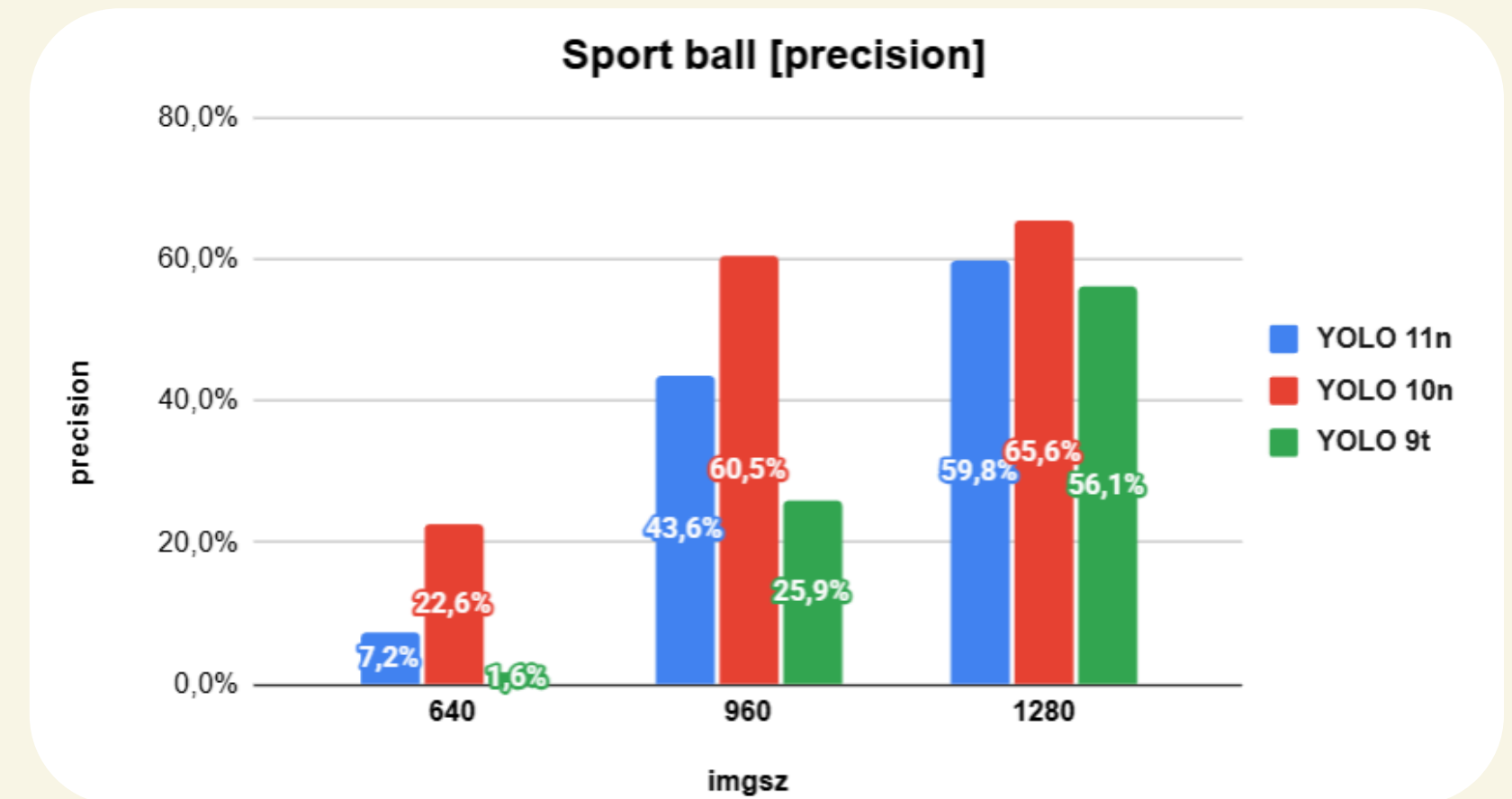
- Test YOLO models pre-trained on COCO dataset
- Establish a baseline performance on padel detection task

2. Evaluate the fine-tuned models

Pre-trained models: Larger imgsz, better precision?



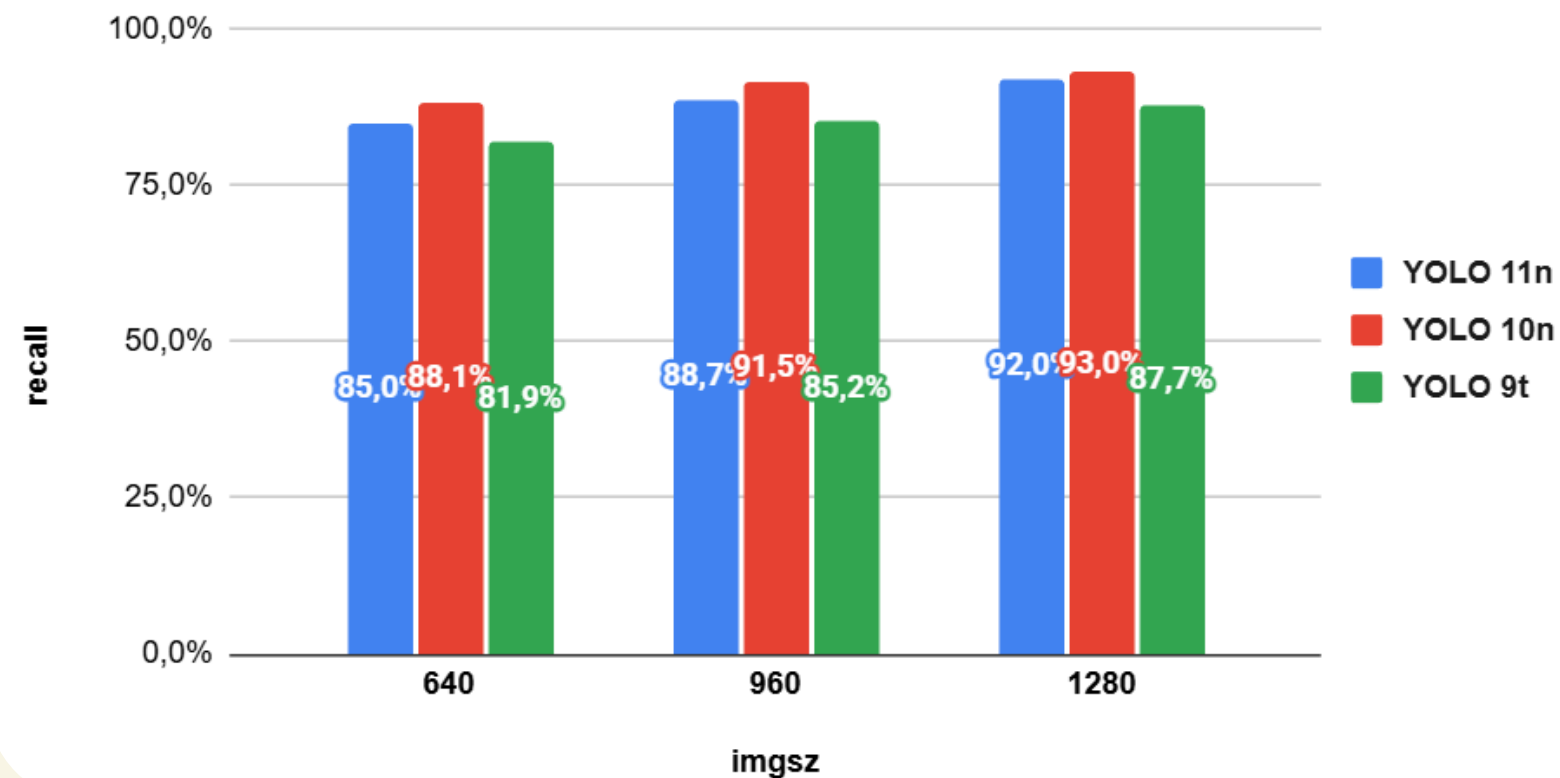
- All models maintain **high precision >85%** at all resolutions
- The difference between the models are minimal; no clear winner emerges



- **Precision increases as imgsz grows**
- At imgsz=1280, YOLO10n achieves the highest precision on sport ball objects

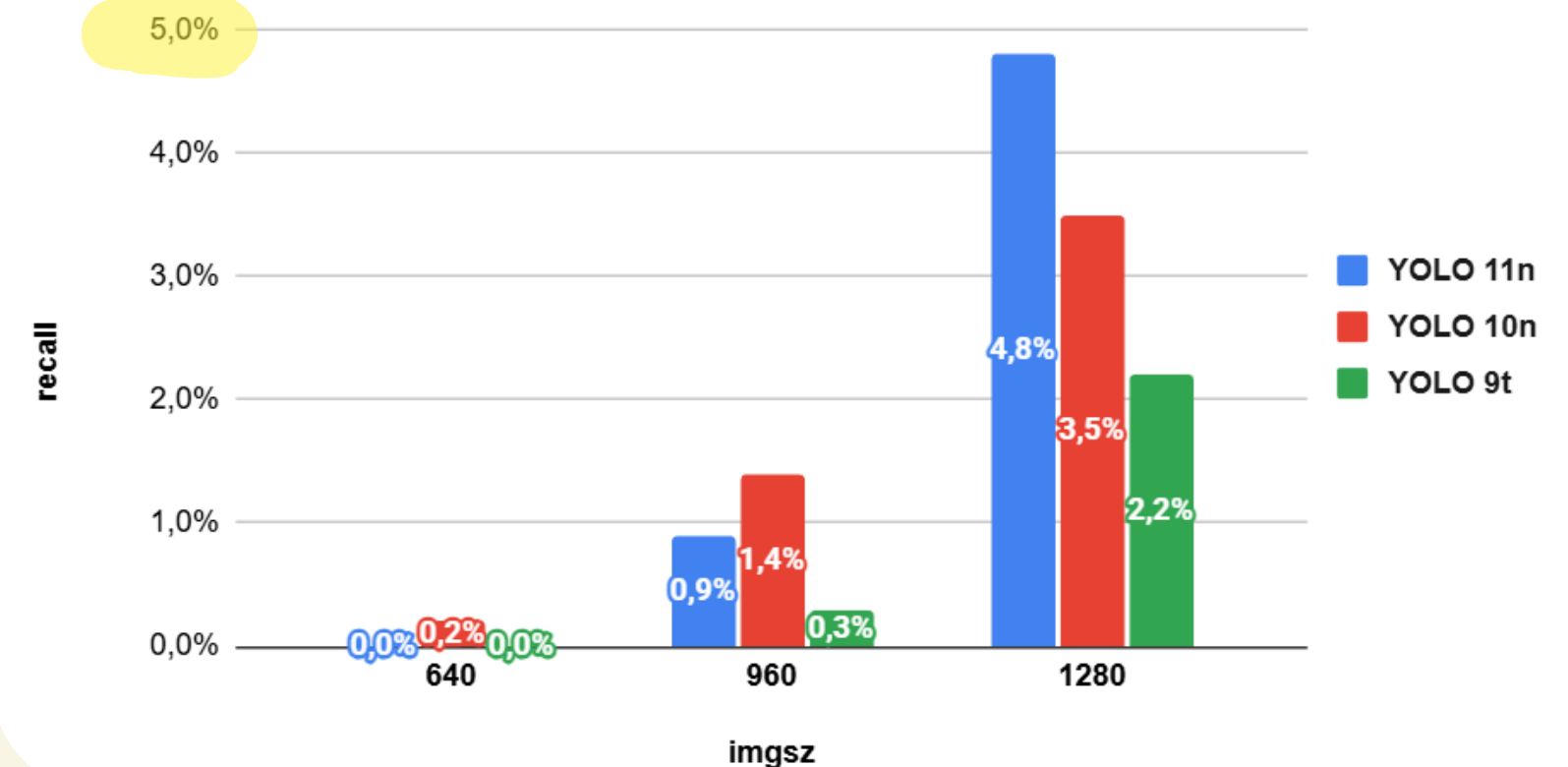
Pre-trained models: Larger imgsz, better recall?

Player [recall]



- High recall at all resolutions (>80%)
- YOLO10n at imgsz=1280 achieves the highest recall
- The recall gain compared to imgsz=640 is about 5%

Sport ball [recall]



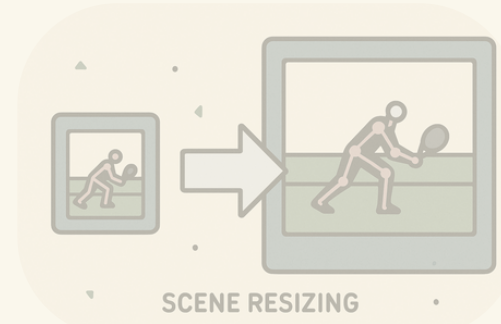
- Results are **very poor** for all models and resolutions (<5%)
- At imgsz=640, recall is insignificant

Models



yolo 11n
yolo 10n
yolo 9t

model comparison



640
960
1280

imgsz hyperparameter

Recall: how many true positives are detected among all actual positives

Precision: how many of the model's positive predictions are correct

metrics

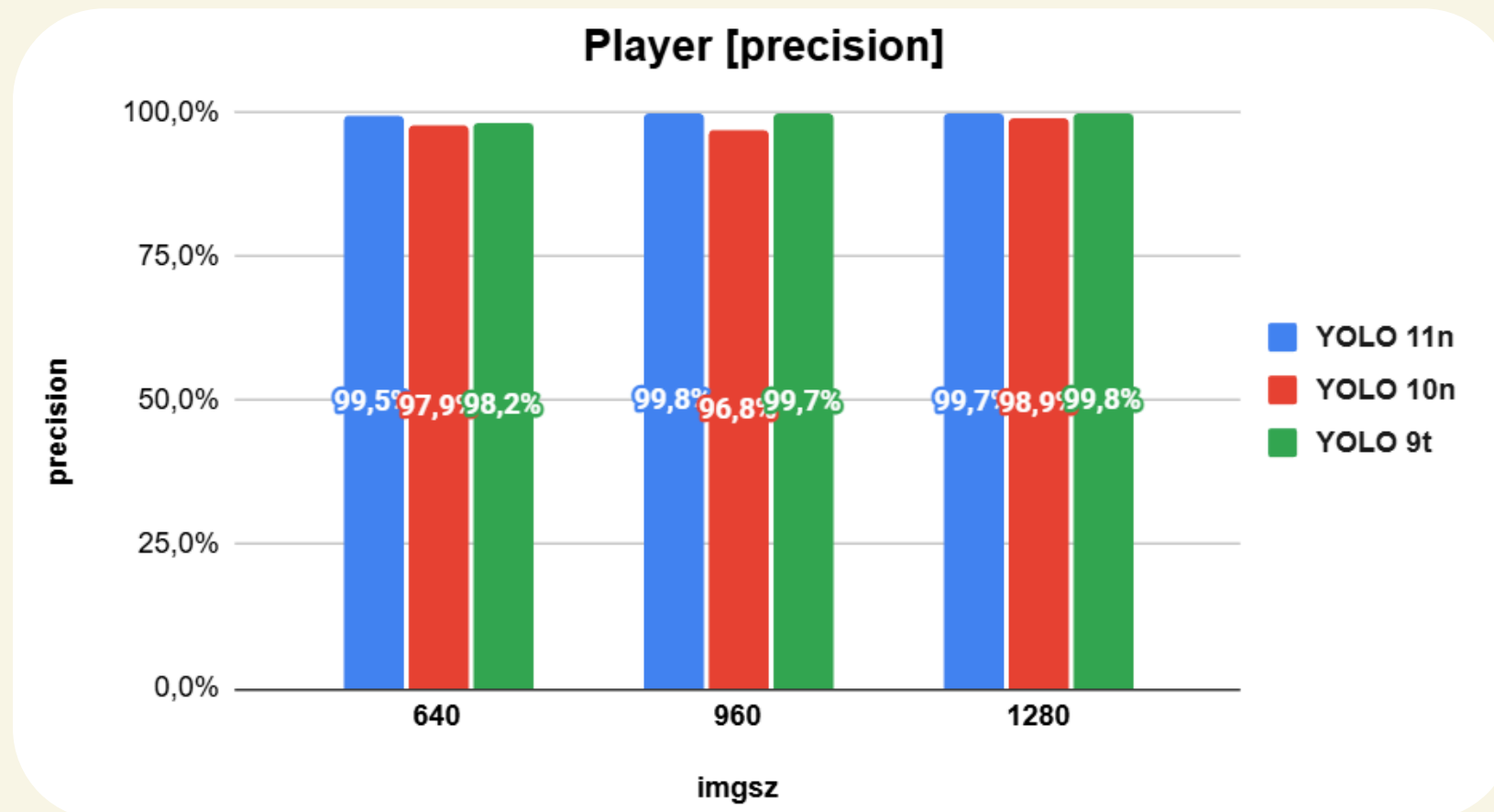
1. Evaluate the pre-trained models

- Test YOLO models pre-trained on COCO dataset
- Establish a baseline performance on padel detection task

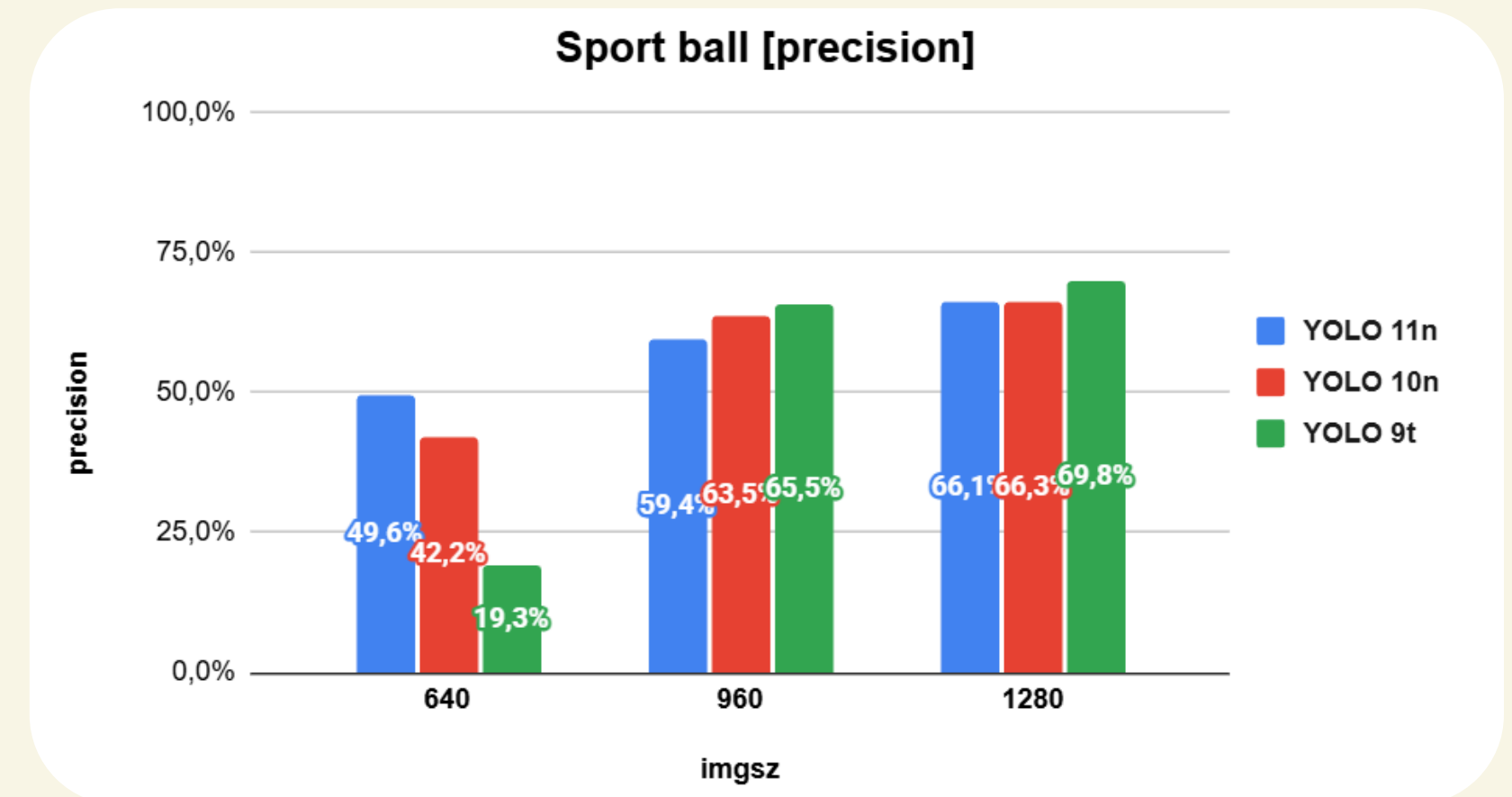
2. Evaluate the fine-tuned models

- Train for 20 epochs
- Adapt the model to the camera viewpoint and match context
- Compare fine-tuned performance against the pre-trained baseline

Fine-tuned models: Larger imgsz, better precision?

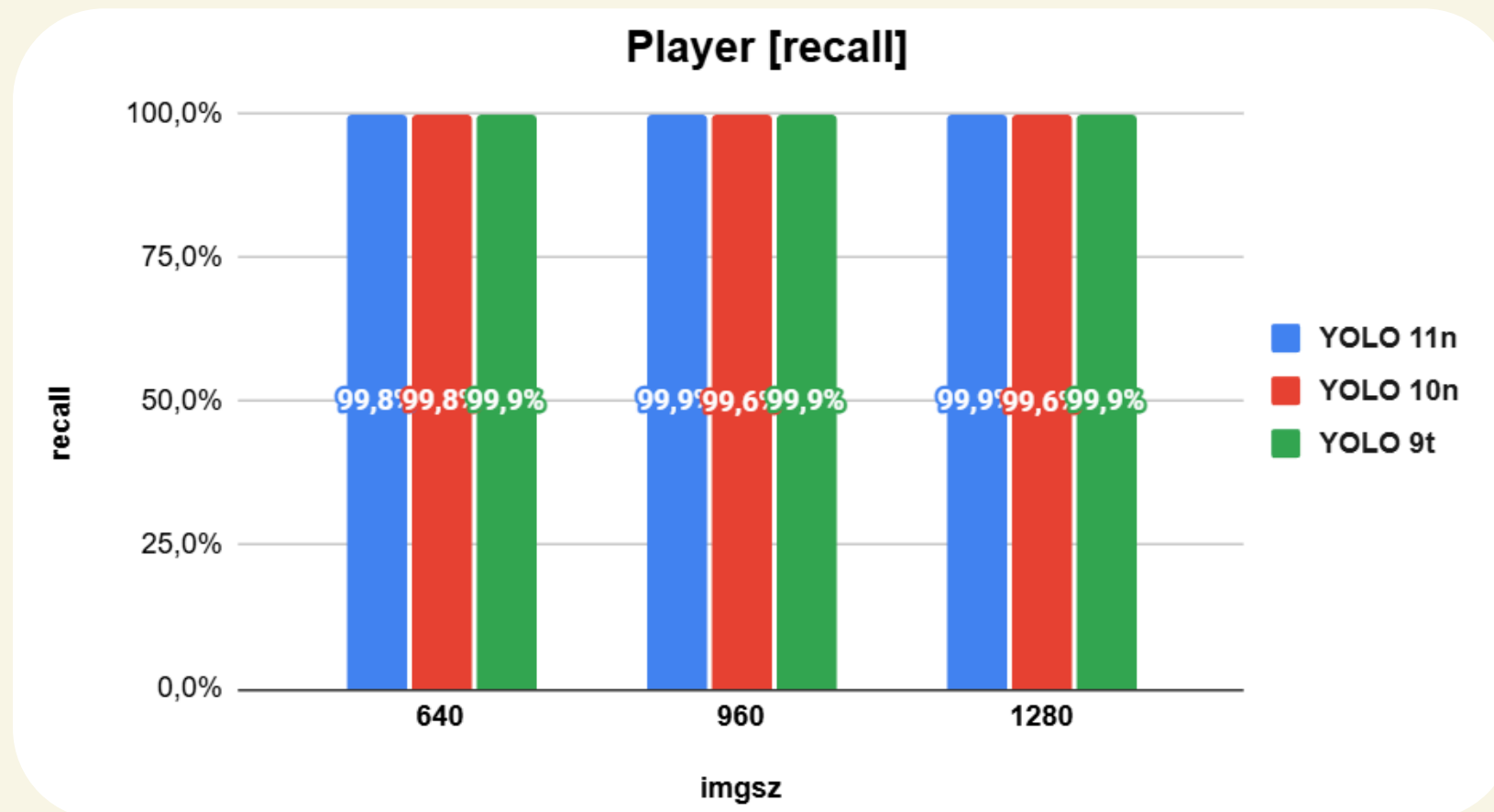


- **Similar performance** for all configurations
- **High precision** for all configurations (>97%)

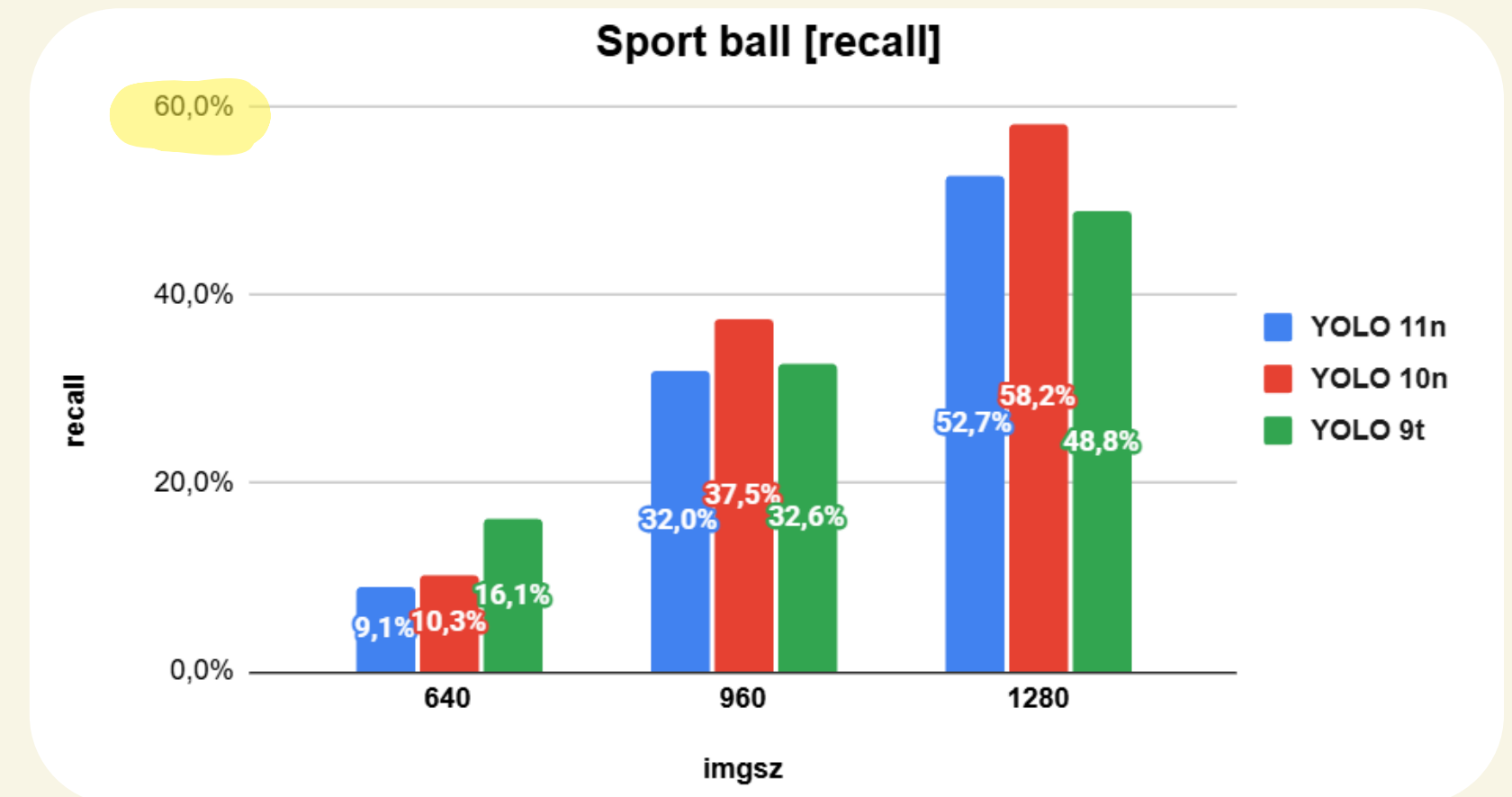


- Precision increases as the image size grows
- With imgsz = 640, the default one, YOLO11n performs best
- Best configuration is Yolo9t with imgsz = 1280

Fine-tuned models: Larger imgsz, better recall?



- Similar performance for all configurations
- High recall for all configurations (>98%)



- Best configuration is Yolo10n with imgsz = 1280
- Recall increases as the image size grows

Fine tuning impact

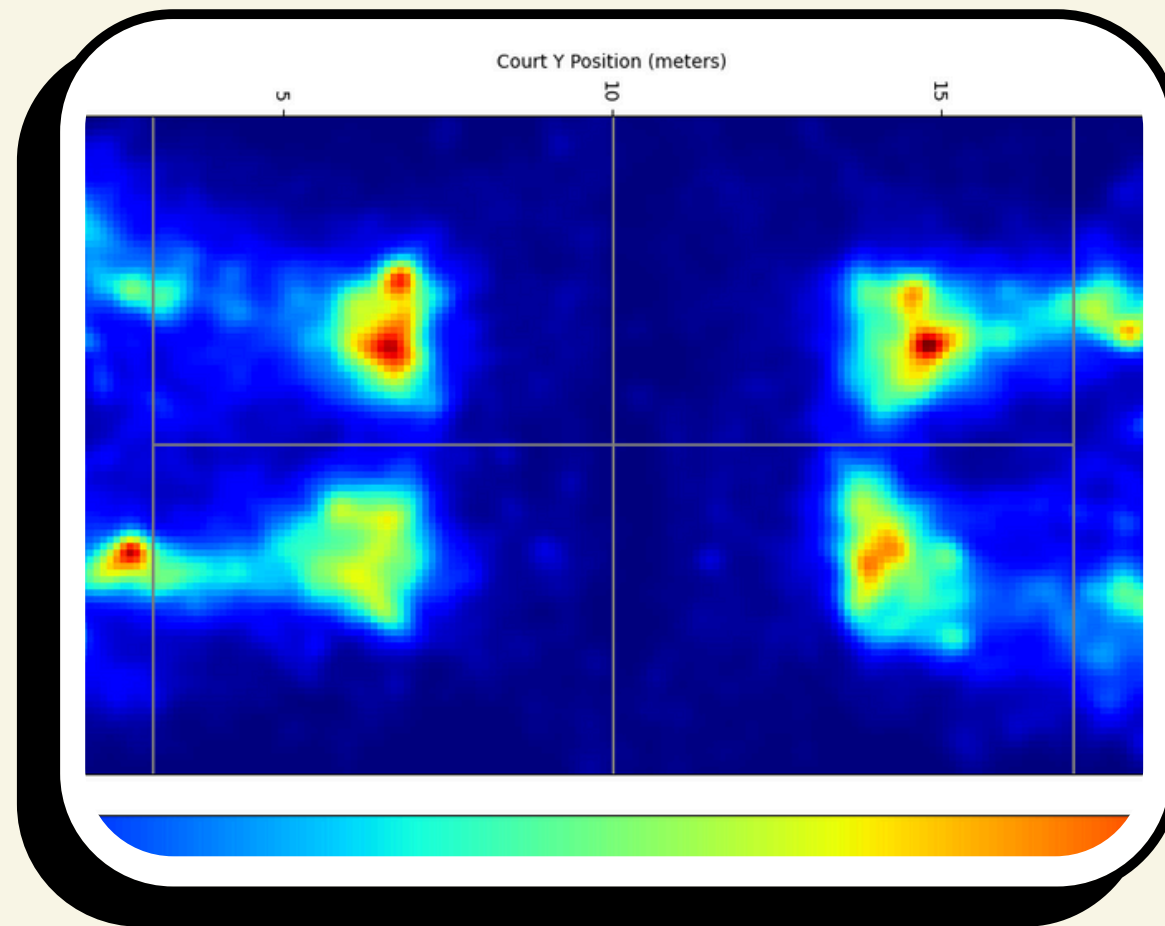
Once the **best configuration** has been selected, we compare performance before and after fine-tuning.

Key observations:

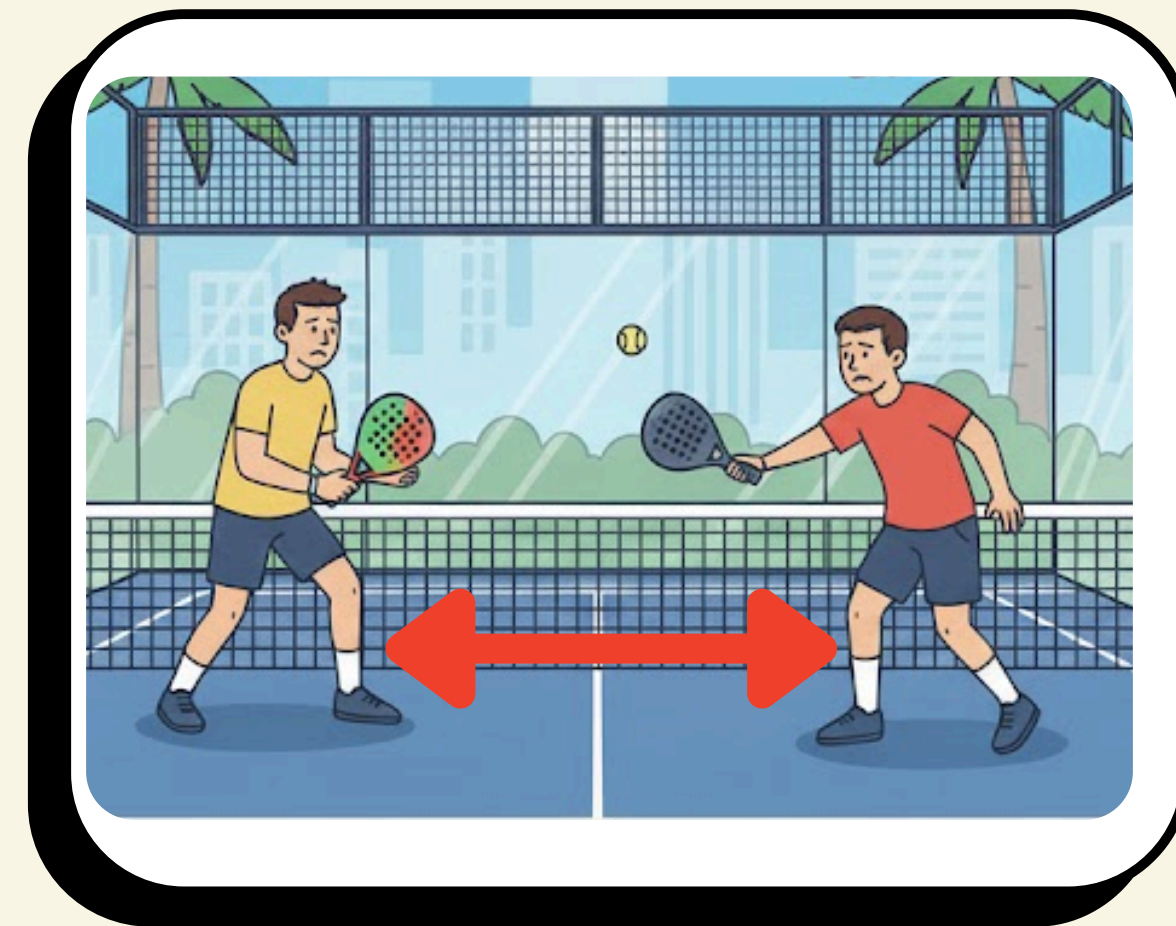
- significant improvement in recall
- greater impact on the sport ball class
- limited benefits in terms of precision

Yolo10n with imgsz=1280			
metrics	class	Δ	avg Δ
precision	person	+13.2%	+7.0%
	sport ball	+0.7%	
recall	person	+6.6%	+30.8%
	sport ball	+54.7%	

Output Statistics



Heatmap



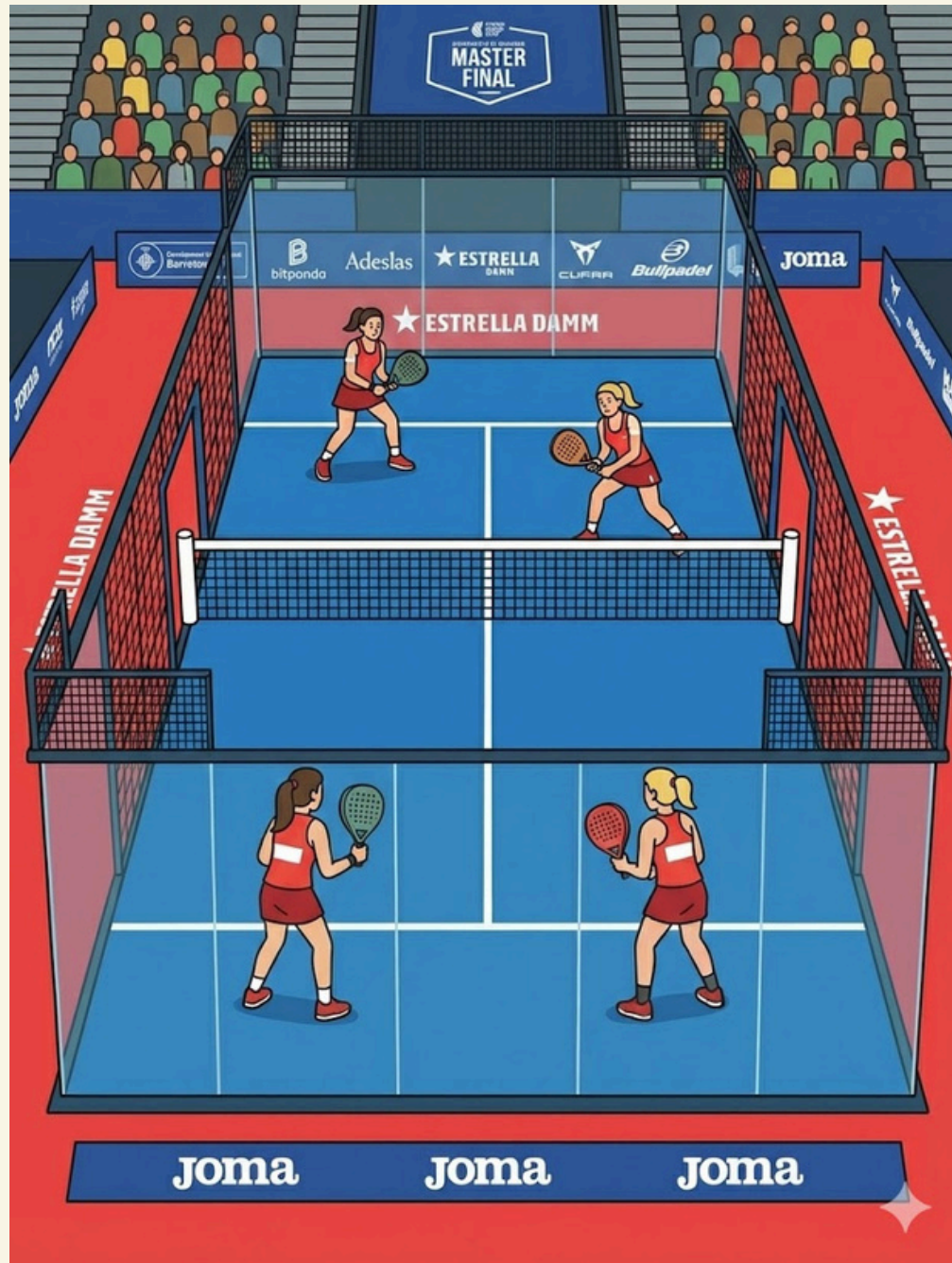
Avg. Distance

Getting the Bounding Boxes

The output of the YOLO-based detector provides **bounding boxes** in pixel coordinates, denoted as $B = (x, y, h, w)$, where (x, y) represents the top-left corner of the bounding box, while w and h represent its width and height



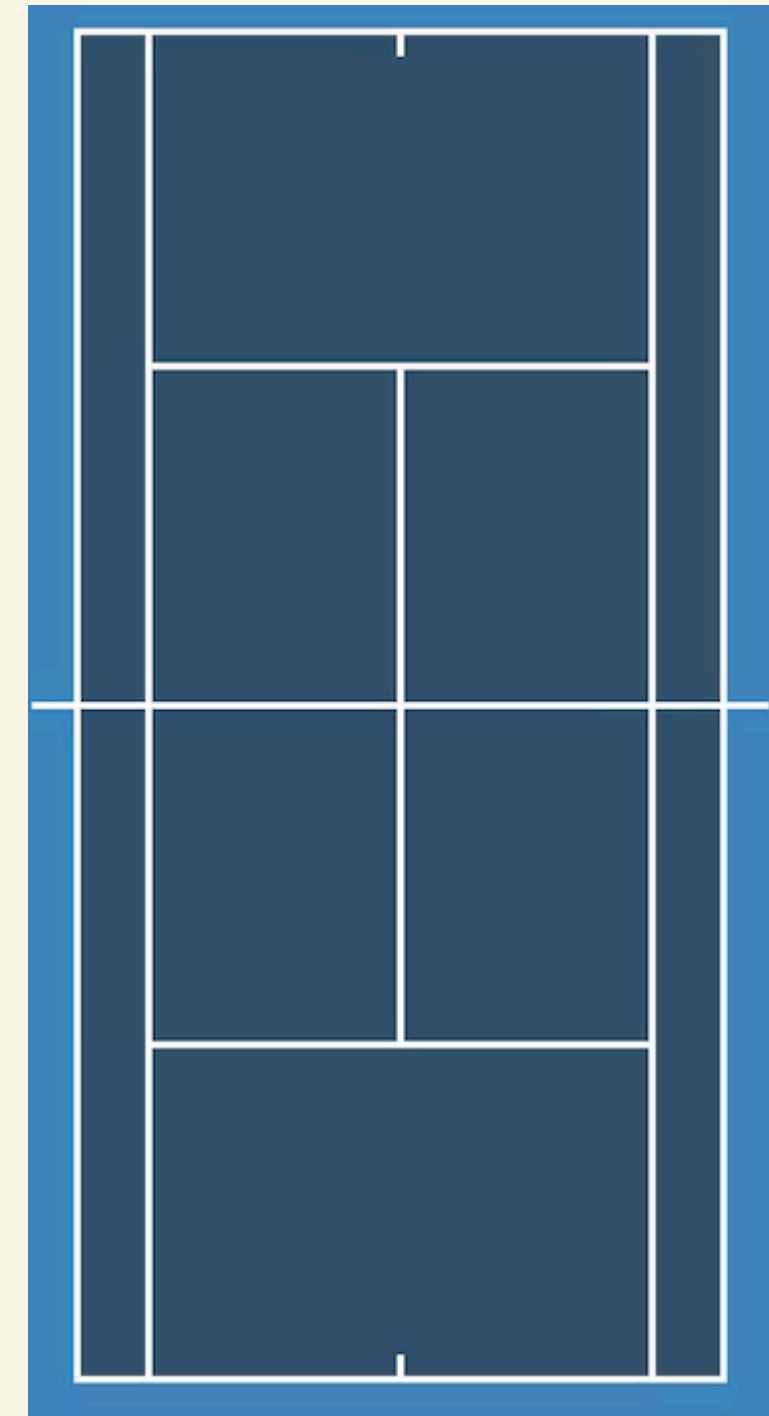
From Pixels to world coordinates



$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$



Homography matrix
computed using **RANSAC**
algorithm



Tracking the players

- First of all players are **initialized**, identities are assigned based on court quadrants.
- The system stores the players' position and velocity at frame t , and uses them to **estimate the players' positions** at frame $t+1$.
- Estimations are compared to detections at frame t , to correctly assign identities; if a detection is missing the estimated position is used in its place.

Player's Velocity at Frame t



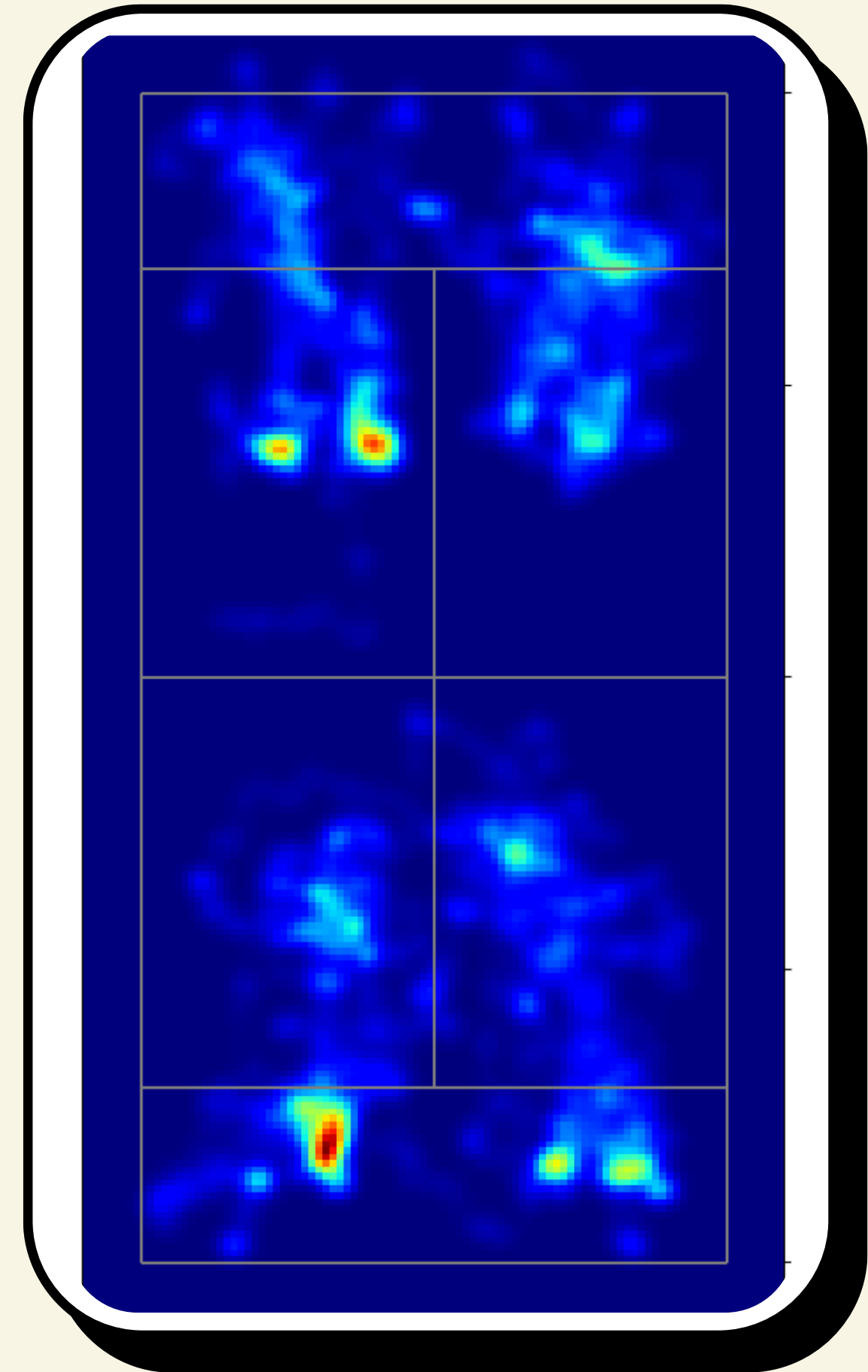
Detected position at Frame t



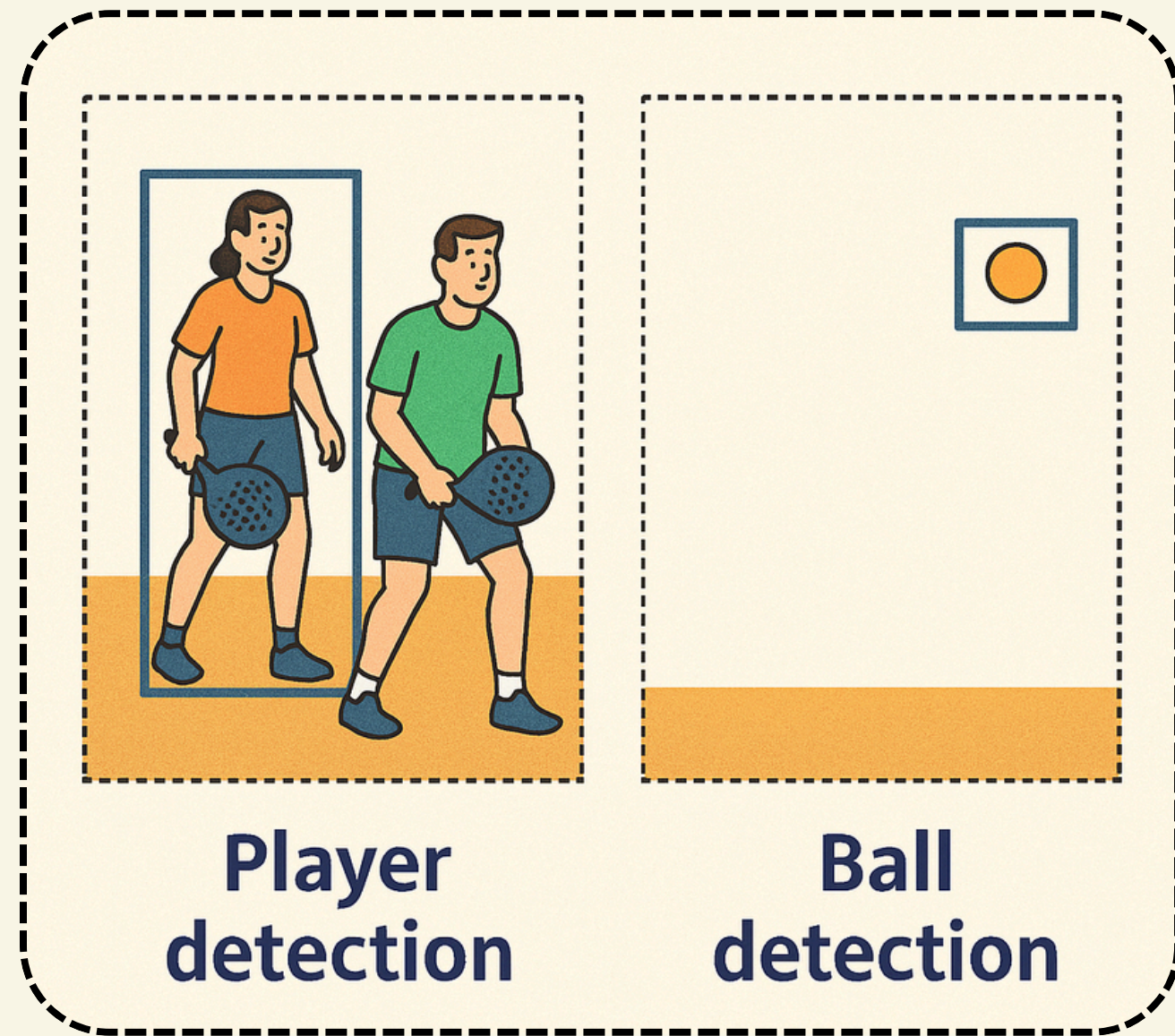
Predicted position at Frame $t+1$

Statistics Visualization

To analyze court coverage, we generate spatial density estimates and world coordinates are aggregated into a 2D histogram. This **heatmap** visualizes the frequency of a player's presence in specific zones, allowing for the identification of tactical tendencies such as baseline defensive play versus net-rushing offensive behavior



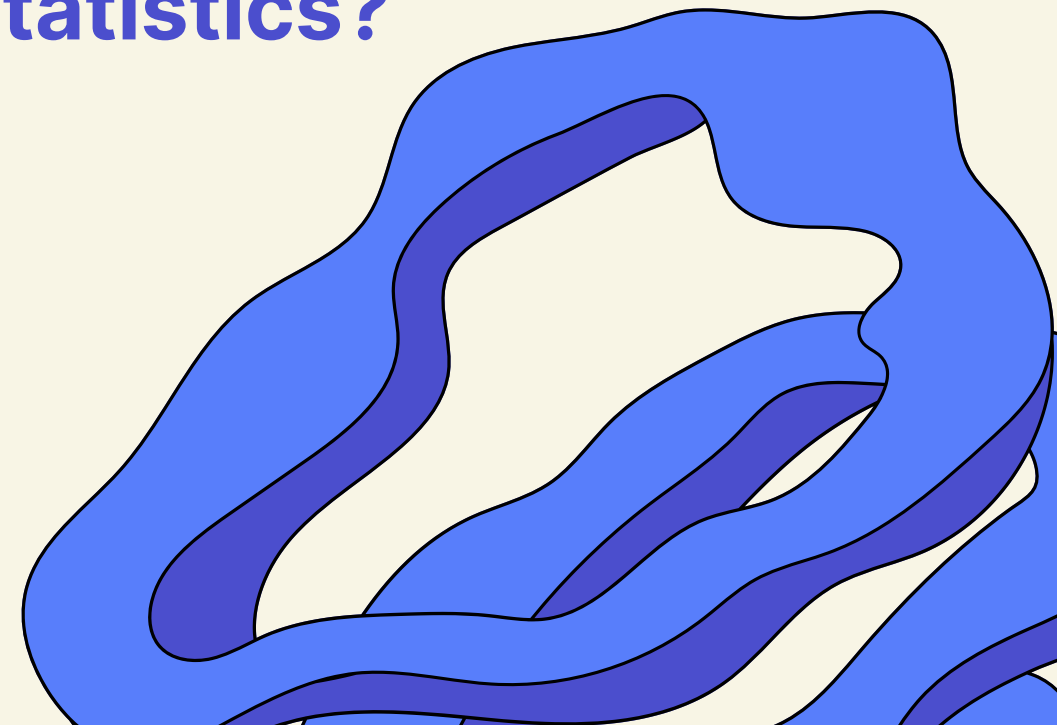
Development of Output Statistics



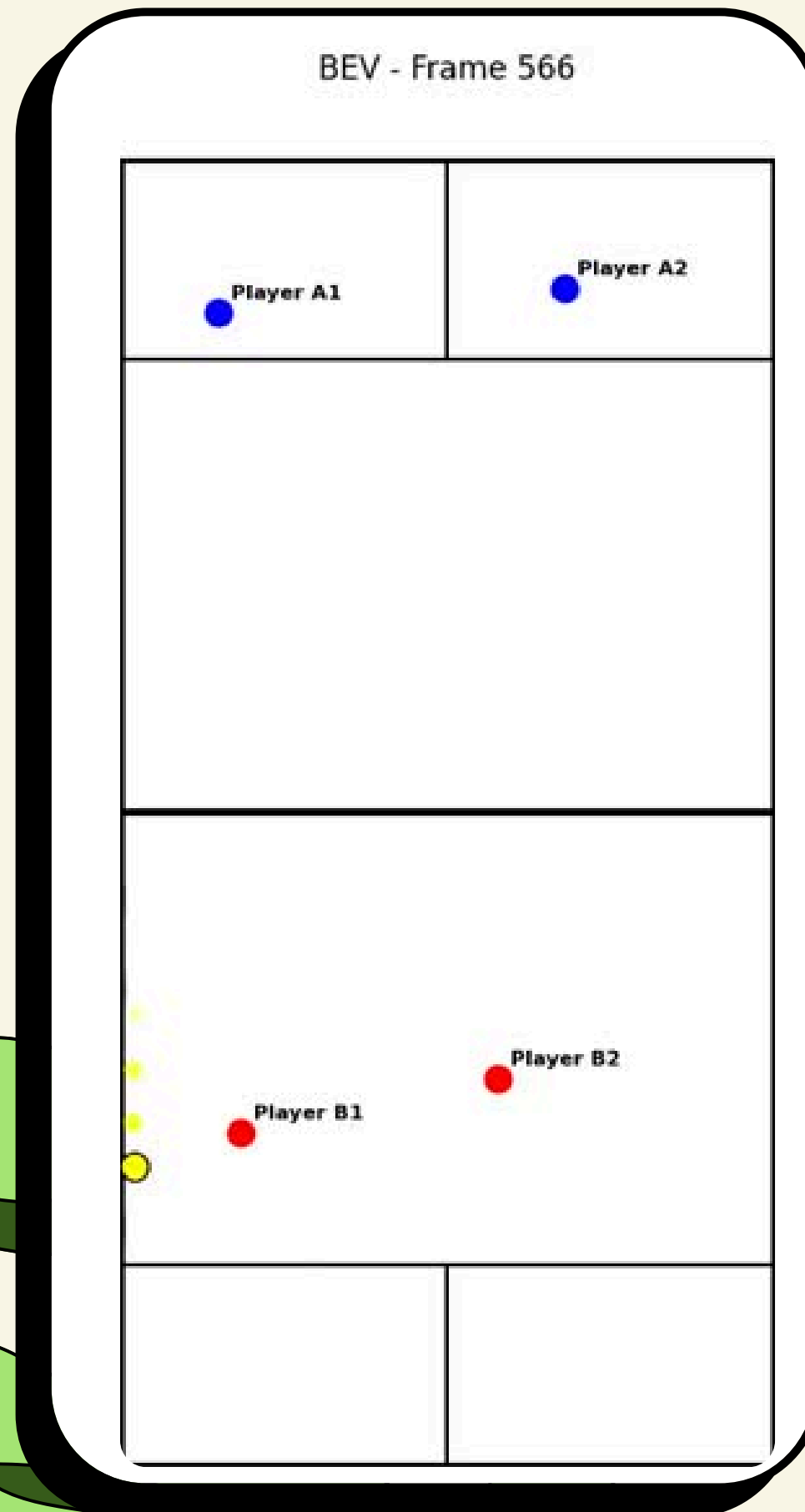
The main aspect of the process of statistic development we can act on is the Bboxes selection from the model's output.

The question we asked ourselves is:

Does using different techniques of bboxes selection actually make a difference in the quality of output statistics?



B.E.V. Output



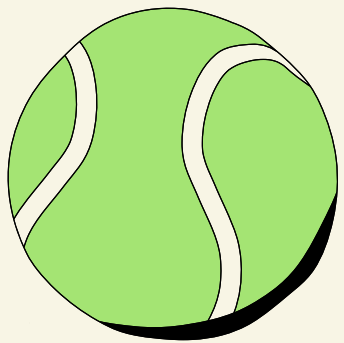
In accordance with our project tutor we decided to produce the Bird's Eye View video as an output of our model. This in the video on the right is the BEV video of the first rally of the female match.

This statistic also highlights **problems and challenges** with **ball location and projection** onto the pitch.

Achievements Summary



● We achieved **near-perfect results** in precision and recall, demonstrating that the model successfully learned to distinguish players from the background.



Precision remained relatively stable, while **recall** showed significant improvement. The model now detects the majority of balls in the padel match.

Although we **lack objective metrics** to evaluate the performance of our output we have made some final considerations.

Filtered process:

- 0.0021% over the whole female match
- 0.0% over the first game

Tracking process:

- 0.0604% over the whole female match
- 0.0585% over the first game

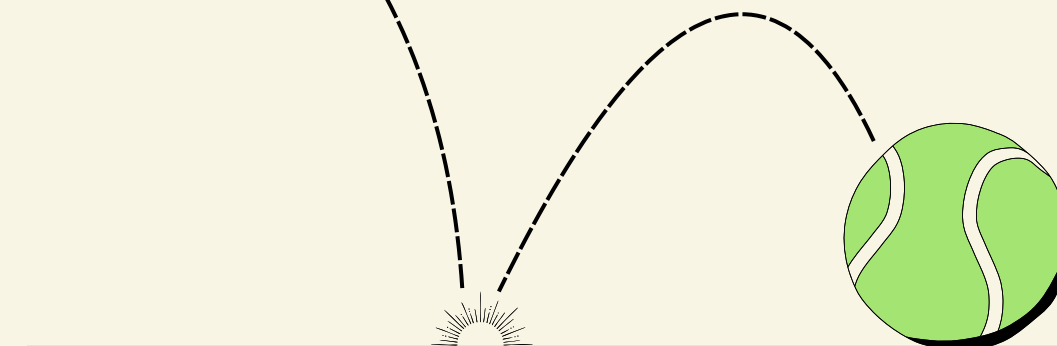
Possible Developments

Ball tracking and Projection:

with an improved dataset and more than one camera angle it would be possible to track the ball and project in onto the field

Shots Statistic: having tracked and projected the ball it would be possible to provide statistics on shots and points

Player Tracking: provided a continuous video it would be possible to track players and handle field switches



Thank you!



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